

A Sharp Physical-to-Affine Remainder for One Critical Endpoint Leg

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July 2026

Abstract

RH-323 propagated the critical endpoint profile through an exact affine probability chain but left the physical curvature and row normalization uncontrolled. We now replace its first tangent leg by the exact finite-noise folded kernel. The physical row has an exact L^1 distance to the full Gaussian centered at the curved quadratic image; this distance is only a pair of state-boundary Gaussian tails and already includes the wrong fold and exact row normalizer. Removing the quadratic displacement then costs order σ . After averaging against the RH-322 entrance row we obtain a fully explicit physical-to-affine joint bound and prove the sharp phase-matched limit

$$\sigma^{-1} \|P_{\sigma,d} - A_d\|_1 \longrightarrow \sqrt{2/\pi} u_c \{d^2 + 1 + d\phi(d)/\Phi(d)\} > 0.$$

Thus the full affine remainder is not exponentially small. Its local $O(\sigma)$ scale is nevertheless smaller than the first-alias target scale. No moving-order composition, second-leg remainder, parity cancellation, or full trace replacement is asserted.

1 The exact physical endpoint leg

Let

$$f(x) = 1 - u_c x^2, \quad r = u_c - 1, \quad \lambda = 2u_c r, \quad \alpha = 2u_c,$$

where

$$u_c = 1.543689012692076\dots, \quad r = 0.543689012692076\dots, \quad \lambda = 1.678573510428322\dots$$

The critical partition is $b = u_c^{-1/2} = 0.8048595351\dots$. For $x, y \in [0, 1]$, the archived folded row is

$$P_\sigma(x, y) = \frac{\phi_\sigma(y - f(x)) + \phi_\sigma(y + f(x))}{Z_\sigma(x)}, \quad \phi_\sigma(t) = \sigma^{-1} \phi(t/\sigma), \quad (1)$$

with $Z_\sigma(x)$ chosen so the row integrates to one.

Use the endpoint and repelling coordinates

$$x = 1 - \sigma v, \quad y = r + \sigma u, \quad I_\sigma = \left[-\frac{r}{\sigma}, \frac{1-r}{\sigma} \right]. \quad (2)$$

Direct substitution gives

$$f(1 - \sigma v) = -r + \sigma c_{\sigma,v}, \quad c_{\sigma,v} = \alpha v - u_c \sigma v^2. \quad (3)$$

The exact rescaled physical row is therefore

$$K_\sigma(v, u) = \frac{\phi(u + 2r/\sigma - c_{\sigma,v}) + \phi(u + c_{\sigma,v})}{Z_\sigma(1 - \sigma v)} \mathbf{1}_{I_\sigma}(u), \quad (4)$$

where, for every physical $0 \leq v \leq \sigma^{-1}$,

$$Z_\sigma(1 - \sigma v) = 1 - \bar{\Phi}\left(\frac{1 - f(1 - \sigma v)}{\sigma}\right) - \bar{\Phi}\left(\frac{1 + f(1 - \sigma v)}{\sigma}\right). \quad (5)$$

Here ϕ and Φ are the standard normal density and distribution function and $\bar{\Phi} = 1 - \Phi$.

On the endpoint branch $1 - \sigma v \geq b$, the second lobe in (4) is the physical folded image. Define its full-line curved Gaussian and the RH-323 tangent row by

$$C_{\sigma,v}(u) = \phi(u + c_{\sigma,v}), \quad A_v(u) = \phi(u + \alpha v). \quad (6)$$

2 Exact row identities

Theorem 2.1 (Fold and normalization collapse to a boundary tail). *If $0 \leq v \leq \sigma^{-1}$ and $1 - \sigma v \geq b$, then*

$$\|K_\sigma(v, \cdot) - C_{\sigma,v}\|_1 = 2 \left\{ \bar{\Phi}\left(\frac{r}{\sigma} - c_{\sigma,v}\right) + \bar{\Phi}\left(\frac{1-r}{\sigma} + c_{\sigma,v}\right) \right\}. \quad (7)$$

Moreover,

$$\|C_{\sigma,v} - A_v\|_1 = 4\Phi\left(\frac{u_c \sigma v^2}{2}\right) - 2. \quad (8)$$

Consequently the physical-to-tangent row distance is at most the sum of (7) and (8).

Proof. The two unnormalized lobes in (1) have total mass one on the positive half-line, so $0 < Z_\sigma \leq 1$. On I_σ , the numerator of (4) contains $C_{\sigma,v}$ and division by $Z_\sigma \leq 1$ only increases it. Thus $K_\sigma(v, u) \geq C_{\sigma,v}(u)$ on I_σ , while the physical row vanishes outside. The positive and negative L^1 parts are equal, and each is the mass of $C_{\sigma,v}$ outside I_σ . Its lower and upper tails are precisely the two terms in (7).

The centers of $C_{\sigma,v}$ and A_v differ by $u_c \sigma v^2$. Two unit Gaussians with center separation t cross at their midpoint and have L^1 distance $4\Phi(|t|/2) - 2$, proving (8). The last assertion is the triangle inequality. $\square$

This identity is stronger than separately bounding the reflected lobe and $Z_\sigma - 1$: their combined effect is exactly the discarded main-lobe boundary mass. The inequality

$$4\Phi(t/2) - 2 \leq \sqrt{\frac{2}{\pi}} t, \quad t \geq 0, \quad (9)$$

will isolate the curvature contribution.

3 The finite-seed physical joint theorem

Let the RH-322 entrance laws be

$$g_a(v) = \frac{\phi(v - a)}{\Phi(a)} \mathbf{1}_{[0, \infty)}(v), \quad (10)$$

$$\tilde{h}_{\sigma,a}(v) = \frac{\phi(v - a) + \phi(2L - v - a)}{\Phi(a) - \Phi(2L - a)} \mathbf{1}_{[0, L]}(v), \quad L = \sigma^{-1}. \quad (11)$$

Define the exact physical one-leg joint law and its affine target by

$$P_{\sigma,a}(v, u) = \tilde{h}_{\sigma,a}(v) K_\sigma(v, u), \quad A_d(v, u) = g_d(v) A_v(u). \quad (12)$$

Fix any

$$0 < \eta < 1 - b, \quad m_\eta = u_c(1 - \eta)^2 - 1 > 0. \quad (13)$$

For reproduction we use the canonical midpoint $\eta = (1 - b)/2 = 0.0975702324\dots$, for which $m_\eta = 0.2571486637\dots$

Put

$$M_2(a) = \int_0^\infty v^2 g_a(v) dv = a^2 + 1 + a \frac{\phi(a)}{\Phi(a)}, \quad c_0 = \sqrt{\frac{2}{\pi}} u_c. \quad (14)$$

Theorem 3.1 (Explicit physical-to-affine joint remainder). *For $\sigma > 0$, $0 \leq a \leq \sigma^{-1}$, and $d \geq 0$,*

$$\begin{aligned} \|P_{\sigma,a} - A_d\|_1 &\leq |a - d| + \frac{2\bar{\Phi}(\sigma^{-1} - a)}{\Phi(a)} + c_0\sigma M_2(a) \\ &\quad + 2 \left\{ \bar{\Phi}(m_\eta/\sigma) + \bar{\Phi}((1 - r)/\sigma) \right\} + \frac{2\bar{\Phi}(\eta/\sigma - a)}{\Phi(a)}. \end{aligned} \quad (15)$$

The same right side bounds the physical U -marginal error. Uniformly for a, d in a fixed compact subset of $[0, \infty)$,

$$\|P_{\sigma,a} - A_d\|_1 \leq |a - d| + C\sigma + Ce^{-c/\sigma^2} \quad (16)$$

for sufficiently small σ , with $C, c > 0$ depending only on the compact set and η .

Proof. Extend $K_\sigma(v, \cdot)$ to all $v \geq 0$ by setting it equal to A_v for $v > L$. Markov lifting preserves the entrance L^1 distance, so RH-322 gives

$$\|\tilde{h}_{\sigma,a}\hat{K}_\sigma - g_a\hat{K}_\sigma\|_1 = \frac{2\bar{\Phi}(L - a)}{\Phi(a)}.$$

The affine lift similarly gives $\|g_a A - g_d A\|_1 = \|g_a - g_d\|_1 \leq |a - d|$.

It remains to average the row error against g_a . On $0 \leq v \leq \eta/\sigma$, the source stays in the negative endpoint branch. The folded image has distance at least m_η from zero, while its distance from the upper state boundary is at least $1 - r$. Hence theorem 2.1 and (9) give the first three remainder terms in (15) after using $\int v^2 g_a = M_2(a)$. On the complement, two probability rows have distance at most two, and

$$\Pr_{g_a}\{V > \eta/\sigma\} = \frac{\bar{\Phi}(\eta/\sigma - a)}{\Phi(a)}.$$

This proves the explicit estimate. Marginalization is an L^1 contraction. Mills' inequality gives (16). $\square$

4 The curvature coefficient is sharp

Theorem 4.1 (Positive first-order obstruction). *For every fixed $d \geq 0$,*

$$\lim_{\sigma \downarrow 0} \frac{\|P_{\sigma,d} - A_d\|_1}{\sigma} = c_0 M_2(d) = \sqrt{\frac{2}{\pi}} u_c \left\{ d^2 + 1 + d \frac{\phi(d)}{\Phi(d)} \right\} > 0. \quad (17)$$

Thus the full physical-to-affine joint remainder is not exponentially small and is not $o(\sigma)$.

Proof. Use the Markov extension from the preceding proof. Replacing the exact finite seed by g_d changes the joint distance by at most $2\bar{\Phi}(\sigma^{-1} - d)/\Phi(d) = o(\sigma)$. Let

$$S_{\sigma,d} = \int_0^\infty g_d(v) \left\{ 4\Phi\left(\frac{u_c \sigma v^2}{2}\right) - 2 \right\} dv.$$

On $v \leq \eta/\sigma$, the reverse triangle inequality and (7) show that the averaged physical row distance differs from $S_{\sigma,d}$ by exponentially small boundary tails. The remaining entrance region also has exponentially small g_d mass.

Pointwise,

$$\frac{4\Phi(u_c\sigma v^2/2) - 2}{\sigma} \rightarrow c_0 v^2,$$

and (9) dominates the quotient by $c_0 v^2$. Dominated convergence and (14) prove (17). $\square$

The reproduction values for the dominant shift integral are

d	$\sigma = 0.05$	0.025	0.0125	$c_0 M_2(d)$
0	1.227163	1.230543	1.231399	1.231686
0.5	1.842222	1.850393	1.852473	1.853170

where the three middle columns are $S_{\sigma,d}/\sigma$. They reproduce the analytic coefficient; they are not a fitted proof.

5 First-alias scale and route firewall

At the natural clock

$$k_\sigma = \frac{\log(1/\sigma)}{2\log\lambda} + O(1),$$

the target radius $R = 1.4$ gives

$$R^{-2k_\sigma} = \Theta(\sigma^\theta), \quad \theta = \frac{\log R}{\log \lambda} = 0.6496301165 \dots < 1. \quad (18)$$

Consequently the phase-matched local remainder satisfies

$$\sigma = o(R^{-2k_\sigma}), \quad k_\sigma \sigma = o(k_\sigma R^{-2k_\sigma}).$$

This is a scalar scale comparison only. It does not prove that a moving-order product has bounded intermediate norms, that all legs satisfy the same estimate, or that weighted trace contributions can be replaced by separate absolute errors.

The implementation evaluates exact normalizers, row tails, Gaussian shift distances, the explicit joint bound, and the sharp coefficient with standard-library quadrature. No exponent is fitted. The result concerns a forward probability density, not the peak-normalized backward Gaussian observable and amplitude coefficient of RH-18.

Remark 5.1 (Claim boundary). The second physical leg, moving-order Duhamel composition, parity layer, neighboring sibling/shell cancellation, branch-return matrix, and joint first-alias trace law remain open. No full-trace replacement or $o(kR^{-2k})$ joint error is proved. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt trace formula or zeta-divisor equality, and does not imply RH.

RH-325 should state a moving-order Duhamel criterion that records retained coordinates, phase transport, per-leg Markov contractions, and the precise sum of local errors before any full-cycle application is attempted.